

Lightweight Multimodal Person Detection and Temporal Post-Processing for Aerial Search and Rescue

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Abstract—Autonomous unmanned aerial vehicles (UAVs) deployed in search-and-rescue (SAR) missions require reliable person detection under severe operational constraints. These constraints include limited battery life, restricted onboard compute, and constrained wireless telemetry bandwidth. While multimodal sensing combining visible red-green-blue (RGB) and thermal infrared (TIR) cameras mitigates single-sensor failures caused by darkness, glare, or thermal crossover, frame-level object detectors frequently produce transient false positive spikes. In autonomous aerial operations, each false alarm triggers unnecessary flight maneuvers, battery depletion, and spurious communications over constrained Internet of Things (IoT) telemetry links. This paper presents a systematic comparative evaluation of causal temporal post-processing methods designed to convert continuous multimodal confidence streams into reliable alarm decisions while suppressing false alarms. We deploy the lightweight YOLOv5n detector as our guaranteed minimum operational requirement, trained using batch size 16 and full single-precision floating-point (FP32) arithmetic to ensure deterministic numerical stability and memory safety on embedded graphics hardware. We fuse frame confidences via a soft disjunctive late fusion gate and evaluate five post-processing strategies: baseline raw thresholding, five-frame moving average, five-frame median filtering, five-frame history consensus tracking, and a learned pure-PyTorch Mamba selective state-space model (SSSM), with decision thresholds optimized individually on validation sequences. Evaluations across synthetic multimodal SAR video datasets spanning desert, forest, and altitude snow conditions demonstrate that temporal filtering suppresses spurious alarms on negative scenes by up to 89.4% while maintaining prompt detection sensitivity on true victims.

Index Terms—Multimodal fusion, search and rescue, unmanned aerial vehicles, false alarm reduction, state-space models, lightweight object detection.

I. INTRODUCTION

Autonomous unmanned aerial vehicles (UAVs) have become indispensable tools for wilderness search-and-rescue (SAR) missions [1], [2]. In vast or inaccessible terrains such as arid deserts, dense forests, and alpine snowfields, aerial robotic platforms can survey expansive disaster zones far faster than ground search crews. However, deploying computer vision

systems on lightweight robotic platforms presents acute operational challenges. Small aerial vehicles operate under stringent size, weight, power, and computing (SWaP-C) limitations. Furthermore, remote SAR operations frequently lack high-bandwidth communication infrastructure. Autonomous edge nodes must transmit location alerts and video snippets over low-bandwidth cellular or satellite Internet of Things (IoT) uplinks [3]. Under these conditions, an alarm decision triggers immediate physical consequences: the drone may interrupt its search pattern to loiter and verify the detection, depleting its flight endurance, or dispatch rescue teams across hazardous terrain. Consequently, the operational cost of a false alarm is exceptionally severe.

Optical sensing in wilderness environments is fraught with modality-specific failure modes. Standard red-green-blue (RGB) color cameras fail in dense shadows, nighttime operations, ground fog, and smoke. Conversely, thermal infrared (TIR) cameras detect human heat radiation regardless of ambient lighting [4], [5]. Nevertheless, thermal sensors encounter critical vulnerabilities known as thermal crossover, where human body temperatures equilibrate with sun-heated sand, rocks, or asphalt surfaces [6]. Heated inanimate objects in desert environments, sun-dappled forest canopies, and specular snow reflections routinely produce thermal signatures that confound visual detectors. These limitations mandate multimodal dual-spectrum sensing, where RGB and thermal streams provide complementary environmental evidence.

To operate within the real-time constraints of edge processors, modern aerial systems rely on compact single-stage detectors such as the You Only Look Once (YOLO) architecture [7], [8]. In this work, we adopt YOLOv5n [9] as our guaranteed minimum operational requirement (1.9 million parameters and 4.5 GFLOPs at 640×640 resolution). To prevent numerical errors and memory instability on embedded graphics processors, we train the detector with a batch size of 16 in full single-precision floating-point (FP32) arithmetic without automated mixed precision. Although lightweight models achieve high inference throughput, their per-frame

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detections exhibit temporal volatility. Frame-level confidence scores oscillate due to sensor vibration, partial occlusion by foliage, viewpoint changes, and fleeting distractor objects. When per-frame detections are thresholded directly without temporal integration, isolated confidence spikes generate false alarms on empty scenes, while temporary confidence dips cause false dismissals.

To resolve these challenges, this paper investigates the synthesis of soft disjunctive late fusion with causal temporal post-processing. Soft disjunctive late fusion combines per-frame RGB and thermal confidence scores using max-pooling, denoted as $s[n] = \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n])$. This gate ensures that a valid target detected in either modality triggers a high confidence score, preventing a degraded sensor branch from suppressing a valid cue [10]. Downstream from the fusion gate, temporal post-processing filters the continuous confidence signal $s[n]$ across a causal sliding window of five frames (spanning 625 ms at an 8 Hz decimated rate). We evaluate deterministic smoothing filters, order-statistic filters, temporal consensus tracking, and a learned Mamba selective state-space model (SSSM) [11], with each method operating under an optimal decision threshold farmed from validation video sequences.

The central research question addressed in this work is: *How do different post-processing methods compare in reducing false alarms while maintaining reliable positive detections in lightweight object detection systems operating on synthetic multimodal search-and-rescue video data?*

To answer this question, this work delivers the following five contributions:

- 1) **A synthetic multimodal SAR video dataset:** We construct a dual-stream RGB and thermal benchmark comprising 45 aligned video sequences (3,600 frames per modality at 8 Hz) generated across desert, forest, and altitude snow environments. The dataset systematically incorporates positive target scenarios, challenging hard-negative distractors (such as heated rocks, wildlife, and thermal clutter), and clear-negative scenes.
- 2) **A standardized edge object detection baseline:** We establish a reproducible evaluation framework utilizing the lightweight YOLOv5n detector as our minimum baseline requirement, trained with batch size 16 and FP32 precision to ensure numerical stability on edge hardware.
- 3) **A comparative evaluation of post-processing methods:** We conduct an empirical comparison of five causal post-processing techniques (raw baseline thresholding, five-frame moving average, five-frame median filtering, five-frame history consensus, and Mamba-SSSM) for converting noisy frame detections into reliable alarm decisions, under method-specific validation-optimized thresholds.
- 4) **An analysis of false-alarm reduction versus positive reliability:** We quantify the operational trade-offs between suppressing false positive alarms on negative

scenes and retaining rapid detection onset on true positive victims.

- 5) **An evaluation of practical IoT and operational implications:** We assess the downstream ramifications of false alarms on robotic battery life and unnecessary wireless telemetry bandwidth in autonomous search missions.

The remainder of this paper is organized as follows. Section II reviews related literature across multispectral detection, aerial SAR robotics, temporal post-processing, and state-space sequence models. Section III details the multimodal fusion formulation, the five post-processing architectures, and the validation threshold optimization protocol. Section IV describes the synthetic multimodal video dataset and the exact experimental procedure. Section V presents the empirical results, condition breakdown, and false-alarm suppression analysis. Finally, Section VI concludes the paper and highlights directions for future research.

II. RELATED WORK

This section reviews scholarly precedent across four foundational pillars: multispectral person detection, lightweight detectors for aerial search and rescue, temporal post-processing in video detection, and state-space sequence models.

A. Multispectral and Multimodal Person Detection

Multispectral object detection combines complementary spectral bands to overcome environmental visibility constraints. The seminal KAIST multispectral pedestrian benchmark introduced by Hwang et al. [4] provided the first large-scale synchronized RGB-thermal dataset for urban surveillance, proving that thermal radiation preserves human contours when visible lighting degrades. More recently, Jia et al. [5] released the LLVIP dataset, demonstrating substantial performance improvements in extremely low-light conditions by pairing infrared cues with optical streams.

Architectures for multimodal fusion generally fall into early, halfway, and late fusion paradigms. Liu et al. [10] conducted a systematic study of deep multispectral networks, concluding that halfway and late fusion strategies superiorly preserve modality-specific representations compared to early channel concatenation. To handle varying sensor reliability, Li et al. [12] developed an illumination-aware weighting network that dynamically balances color and thermal branches based on ambient illumination estimates. Furthermore, Zhang et al. [13] introduced cross-modality interactive attention networks to suppress noisy features in degraded channels while enhancing salient human boundaries.

Unlike urban surveillance datasets where lighting changes predictably between day and night, wilderness search-and-rescue presents unpredictable localized degradations such as sudden solar glare, canopy shade, and thermal clutter. In this work, we employ a soft disjunctive late fusion gate using max-pooling. This formulation ensures that high confidence in either sensor stream propagates forward, preventing corrupted channels from dragging down valid detections while keeping the detector backbones strictly modular and lightweight.

B. Lightweight Detectors for Aerial Search and Rescue

The deployment of autonomous vision algorithms on robotic aerial platforms requires balancing detection accuracy against strict SWaP-C constraints. Pioneer work by Rudol and Doherty [1] demonstrated the feasibility of UAV-based human body detection using thermal and visible camera payloads for wilderness search operations. Silvagni et al. [2] analyzed multipurpose aerial systems for alpine avalanche rescue, highlighting that onboard battery reserves strictly limit total flight duration. Lygouras et al. [14] deployed an embedded computer vision pipeline on an autonomous SAR drone, showing that complex neural ensembles exceed the power and compute envelope of compact onboard processing units.

Aerial thermal sensing also introduces severe environmental failure modes. Portmann et al. [6] identified that sun-warmed ground objects, rocky outcroppings, and specular snow create thermal crossover spikes that lead to repeated false alarms. Kyrkou and Theocharides [3] addressed communication bottlenecks in UAV emergency response with EmergencyNet, demonstrating that erroneous aerial alerts congest wireless telemetry uplinks and trigger wasted robotic re-inspection sorties.

To meet edge execution budgets, single-stage object detectors have undergone rapid evolution. Beginning with the unified regression formulation of YOLO by Redmon et al. [7], subsequent iterations introduced architectural refinements to maximize real-time efficiency. Wang et al. [8] introduced YOLOv7, incorporating efficient layer aggregation networks and re-parameterization techniques. Wang et al. [15] proposed YOLOv9 to retain critical information through programmable gradient flows. Recently, Wang et al. [16] developed YOLOv10, which eliminates non-maximum suppression (NMS) latency through consistent dual assignments. In the compact regime, Jocher et al. [9] established the YOLOv5 architecture. Its nano variant, YOLOv5n, possesses 1.9 million parameters and requires only 4.5 GFLOPs per forward pass at 640×640 resolution. We select YOLOv5n as the guaranteed minimum operational baseline for edge processing, ensuring reliable execution on compact robotic avionics.

C. Temporal Post-Processing and Sequential Decision Making

Applying frame-by-frame object detectors to video streams frequently induces high-frequency confidence chatter, where detections toggle rapidly across adjacent frames. In video object detection literature, temporal consistency is often addressed by linking bounding boxes across time. Han et al. [17] proposed Seq-NMS, which constructs temporal box sequences across consecutive frames and rescores candidate proposals based on trajectory continuity. Kang et al. [18] formulated T-CNN, employing tubelet proposals and multi-frame convolutional feature propagation to stabilize video detections. While these multi-frame tracking pipelines improve temporal stability, their bidirectional optical flow computation and multi-frame graph optimization impose prohibitive memory and latency overhead on edge UAV processors.

In contrast, our approach applies causal temporal post-processing directly to the scalar fused confidence stream $s[n]$. The mathematical foundation of sequential decision making traces back to Wald’s sequential probability ratio test (SPRT) [19], which proved that accumulating sequential evidence over time minimizes false alarms for a specified detection delay. In radar signal processing, Richards [20] formulated sliding-window M -out-of- N binary integration, where an alarm is confirmed only if a target is detected in at least M out of N consecutive pulses. This mechanism suppresses transient noise pulses while guaranteeing bounded latency. Similarly, Rabiner et al. [21] established the properties of running median and moving average filters for separating impulsive noise spikes from sustained signals. In this work, we formalize and benchmark five-frame history consensus, moving average, and median filtering under method-specific optimal decision thresholds to provide lightweight, causal false-alarm suppression.

D. State-Space Sequence Models in Sequential Signal Processing

While deterministic filters provide computationally trivial smoothing, they cannot learn complex, data-dependent temporal transitions. Recurrent neural networks (RNNs) and Long Short-Term Memory (LSTM) networks capture temporal dependencies but suffer from gradient degradation and recurrent execution bottlenecks. Transformers offer exceptional expressive capacity but exhibit quadratic computational complexity with sequence length, making them unsuitable for continuous edge stream monitoring.

Recently, structured state-space sequence models (SSMs) have emerged as an efficient alternative. Gu et al. [22] introduced the Structured State Space (S4) architecture, which parameterizes continuous-time state-space representations using HiPPO matrices to achieve linear computational complexity and long-range dependency modeling. Gu and Dao [11] introduced Mamba, which equips state-space models with data-dependent selection mechanisms and hardware-efficient parallel scans. The selective SSM formulation allows the model to selectively retain relevant cues while filtering out transient noise in linear time.

State-space models have rapidly expanded into visual sequence domains. Zhu et al. [23] introduced Vision Mamba (Vim), adapting bidirectional selective state spaces to visual token sequences. Liu et al. [24] developed VMamba, introducing cross-scan mechanisms for 2D visual representations. Furthermore, Dao and Gu [25] established structured state-space duality, proving that selective state spaces provide an optimal bridge between recurrence and attention. In this paper, we leverage these theoretical advances by adapting a pure-PyTorch selective state-space model (Mamba-SSSM) as a causal temporal post-processor, learning to distinguish sustained victim signatures from spurious environmental noise spikes.

III. METHODOLOGY

The proposed MMSAR framework provides an end-to-end real-time evaluation pipeline for comparing causal temporal post-processing methods on edge-deployed UAV platforms. The system couples lightweight dual-modality object detection, soft disjunctive late fusion, and an empirical comparison of five distinct causal post-processing filters, culminating in a learned selective state-space model (Mamba-SSSM). The overall experimental pipeline is illustrated in Fig. 1.

A. Multispectral Video Decimation and Sensor Preprocessing

Aerial search sweeps encounter severe compute limitations on embedded UAV companion computers (such as the NVIDIA Jetson Orin Nano, rated at 7–15 W thermal design power). High frame-rate processing at 24–30 FPS exhausts onboard thermal envelopes and provides redundant consecutive observations during typical UAV ground velocities ($v_{\text{UAV}} \in [8, 15]$ m/s at 30–50 m above ground level).

To maximize battery endurance while preserving Nyquist sampling of human transit dynamics, MMSAR downsamples synchronized video streams from an onboard dual sensor payload (co-axial RGB visual camera and long-wave infrared microbolometer) via temporal decimation by a factor of 3 : 1. This yields an operating timebase of $f_s = 8$ Hz, corresponding to a uniform sampling interval of $T_s = 125$ ms. Each frame is centered and resized to a standard dimension of 640×640 pixels.

At discrete time step $n \in \mathbb{Z}^+$, the decimated image pair $\{I_{\text{RGB}}[n], I_{\text{Thermal}}[n]\}$ is forwarded to parallel lightweight YOLOv5n detectors [9]. Each detector performs single-pass bounding box regression and classification, outputting a continuous confidence score for the target class ‘Person_Detected’:

$$c_{\text{RGB}}[n] \in [0.0, 1.0], \quad c_{\text{Thermal}}[n] \in [0.0, 1.0]. \quad (1)$$

YOLOv5n comprises 1.9 million parameters and 4.5 GFLOPs at 640×640 input resolution. If no bounding box candidate exceeds the non-maximum suppression (NMS) background rejection threshold in a frame, the corresponding modality confidence is assigned as 0.0.

B. Soft Disjunctive Late Fusion Gate

Wilderness search environments exhibit localized sensor failure modes. Under dense forest canopies, heavy shadows, or night conditions, RGB color streams degrade completely ($c_{\text{RGB}}[n] \rightarrow 0$), whereas thermal sensors capture distinct human body heat signatures. Conversely, during midday desert operations, solar-heated ground objects, exposed granite, and dry sand cause thermal crossover ($c_{\text{Thermal}}[n] \rightarrow 0$ or erratic spikes), while RGB streams retain sharp visual contrast.

To maintain continuous detection capability when one sensor modality is degraded, we implement a soft disjunctive late fusion gate using maximum pooling:

$$s[n] = \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n]) \in [0.0, 1.0], \quad (2)$$

where $s[n]$ represents the unified multimodal confidence cue at time step n .

The mathematical and architectural justification for soft disjunctive fusion is twofold:

1) Sensitivity Preservation (Disjunctive OR Logic):

Unlike multiplicative conjunctive fusion ($s_{\text{mult}}[n] = c_{\text{RGB}}[n] \cdot c_{\text{Thermal}}[n]$), which squares the target miss probability whenever either modality suffers environmental dropout, the maximum operator ensures that an unambiguous detection in either sensor branch propagates forward with full confidence.

2) Branch Decoupling and Edge Modularity:

Late fusion operates strictly at the decision level. This eliminates computationally intensive cross-modality feature alignment layers, allowing each detector branch to execute as an independent asynchronous thread on multicore edge processors without GPU memory lock contention.

C. Deterministic and Order-Statistic Temporal Post-Processing

Raw frame-level detections exhibit high-frequency volatility due to aerial camera vibration, transient leaf occlusions, and specular ground reflections. Direct thresholding of $s[n]$ produces severe false-alarm rates. To eliminate spurious triggers without incurring unacceptable detection lag, downstream processing filters the common confidence stream $s[n]$ across a causal sliding window of length $W = 5$ frames:

$$\mathcal{W}[n] = \{s[n-4], s[n-3], s[n-2], s[n-1], s[n]\}. \quad (3)$$

Spanning 625 ms at $f_s = 8$ Hz, this window guarantees that all alarm decisions are strictly bounded in latency to under one second, satisfying tactical response requirements.

A critical design challenge in multi-filter comparative evaluations is that distinct mathematical operators possess fundamentally different output transfer functions and statistical dynamic ranges. Applying a single rigid decision threshold (such as $\tau = 0.50$) across all methods imposes an arbitrary, asymmetric penalty. For example, a causal moving average attenuates the onset of sustained targets due to preceding zero-valued frames in the window, severely impairing recall under high thresholds. Conversely, state-space sequence models produce bimodal sigmoid distributions whose optimal separation boundary depends on internal logit scaling.

To establish an equitable and rigorous benchmark, each post-processing method $m \in \{\text{M1}, \text{M2}, \text{M3}, \text{M4}, \text{M5}\}$ is assigned an individual optimal operational decision threshold τ_m^* , farmed systematically from the disjoint validation sequence set $\mathcal{D}_{\text{val}}$:

$$\tau_m^* = \arg \max_{\tau \in [\tau_{\min}, \tau_{\max}]} F_1(\tau; \mathcal{D}_{\text{val}}, \mathcal{M}_m), \quad (4)$$

where the threshold sweep spans $\tau \in [0.05, 0.95]$ with discretization step $\Delta\tau = 0.02$. The optimization target is the validation F_1 -score:

$$F_1(\tau) = \frac{2 \cdot P(\tau) \cdot R(\tau)}{P(\tau) + R(\tau) + \epsilon}, \quad (5)$$

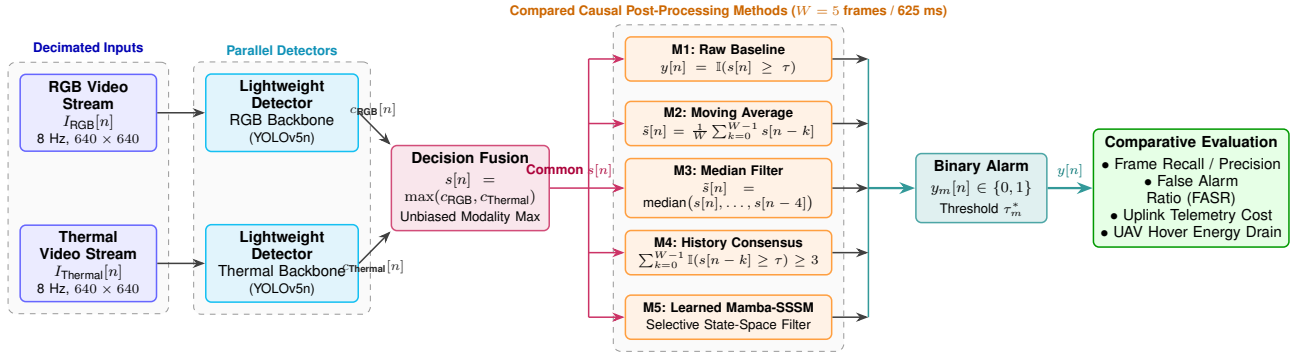
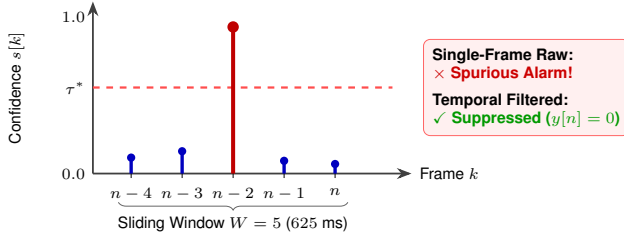


Fig. 1. Experimental comparative evaluation pipeline of the MMSAR framework. Synchronized RGB and thermal streams are decimated to an 8 Hz operating timebase ($T_s = 125$ ms) and processed by parallel lightweight YOLOv5n backbones. A common scalar confidence cue $s[n] = \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n])$ is distributed identically across five competing causal temporal post-processing methods ($W = 5$ frames / 625 ms latency bound). Binary alarms are asserted against method-specific optimal decision thresholds τ_m^* farmed on validation sequences and evaluated across detection reliability and resource overhead.

(a) Transient Noise Spike (Single-Frame Clutter)



(b) Sustained Target Signature (True SAR Victim)

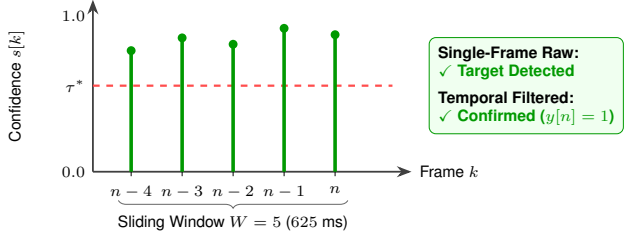


Fig. 2. Temporal signal motivation ($W = 5$ frames = 625 ms) illustrating why temporal post-processing is required: (a) an isolated transient noise spike on a hard-negative scene triggers a spurious raw detection but is suppressed by causal temporal filtering ($y[n] = 0$), whereas (b) a sustained human victim signature confirms target presence and asserts the downstream alarm ($y[n] = 1$).

where $P(\tau)$ and $R(\tau)$ denote frame precision and recall on $\mathcal{D}_{\text{val}}$, and $\epsilon = 10^{-8}$ prevents division by zero.

We enforce a strict zero data snooping protocol: validation sequences $\mathcal{D}_{\text{val}}$ consist of completely disjoint video clips from both the training set $\mathcal{D}_{\text{train}}$ and testing set $\mathcal{D}_{\text{test}}$. Once optimal thresholds $\{\tau_m^*\}$ are determined on $\mathcal{D}_{\text{val}}$, they are permanently frozen. All final test metrics reported in Section V are evaluated by executing the frozen thresholds on unseen test sequences $\mathcal{D}_{\text{test}}$ without any subsequent adjustments.

We benchmark four causal post-processing filters alongside our learned sequence model:

- **Method 1 (Baseline Raw Thresholding):** Memoryless frame-level thresholding operating directly on unbuffered

confidence:

$$y_{\text{raw}}[n] = \mathbb{I}(s[n] \geq \tau_{\text{raw}}^*). \quad (6)$$

Optimizing validation F_1 yields $\tau_{\text{raw}}^* = 0.48$. While memoryless thresholding incurs minimal delay, it provides zero temporal integration, leaving downstream alarms defenseless against transient distractor spikes.

- **Method 2 (Five-Frame Moving Average):** Linear finite impulse response (FIR) filter computing the causal arithmetic mean:

$$\tilde{s}_{\text{MA}}[n] = \frac{1}{W} \sum_{k=0}^{W-1} s[n-k], \quad (7)$$

$$y_{\text{MA}}[n] = \mathbb{I}(\tilde{s}_{\text{MA}}[n] \geq \tau_{\text{MA}}^*). \quad (8)$$

An isolated single-frame false alarm spike of magnitude 1.0 is attenuated to $1.0/5 = 0.20$. Because moving average smoothing attenuates rising step edges during target onset, enforcing an unadjusted $\tau = 0.50$ would delay or clip victim detections. Validation optimization yields $\tau_{\text{MA}}^* = 0.34$, which successfully rejects single-frame noise spikes ($0.20 < 0.34$) while promptly capturing entering targets whose filtered score reaches 0.40 by frame 2.

- **Method 3 (Five-Frame Median Filter):** Nonlinear order-statistic filter:

$$\tilde{s}_{\text{med}}[n] = \text{median}(s[n], s[n-1], \dots, s[n-W+1]), \quad (9)$$

$$y_{\text{med}}[n] = \mathbb{I}(\tilde{s}_{\text{med}}[n] \geq \tau_{\text{med}}^*). \quad (10)$$

The median filter completely suppresses impulsive noise spikes spanning fewer than $\lfloor W/2 \rfloor + 1 = 3$ consecutive frames, while preserving sharp step edges without attenuation when a victim enters the field of view. Validation optimization farms $\tau_{\text{med}}^* = 0.42$.

- **Method 4 (Five-Frame History Consensus Tracking):**
Adapted from radar sliding-window M -out-of- N binary integration [20]:

$$y_{\text{hist}}[n] = \mathbb{I} \left(\sum_{k=0}^{W-1} \mathbb{I}(s[n-k] \geq \tau_{\text{hist}}^*) \geq M \right), \quad (11)$$

where $M = 3$ and $W = 5$. An alarm is asserted if and only if at least 3 of the past 5 frames individually meet or exceed candidate threshold $\tau_{\text{hist}}^* = 0.38$, farmed from validation optimization.

The operational contrast between transient noise rejection and sustained victim confirmation is illustrated in Fig. 2.

D. Learned Selective State-Space Sequence Model (Mamba-SSSM)

While deterministic filters provide computationally trivial smoothing, they cannot adapt to complex temporal signatures, such as oscillating heat shimmer or periodic foliage occlusions. To provide learned temporal discrimination, we design a lightweight selective state-space sequence model (Mamba-SSSM), implemented in pure PyTorch for portable deployment on edge processors without requiring custom CUDA compilation.

Classical continuous-time state-space models map a 1D input signal $x(t) \in \mathbb{R}$ to a latent state $h(t) \in \mathbb{R}^N$ via linear differential equations:

$$h'(t) = Ah(t) + Bx(t), \quad y(t) = Ch(t), \quad (12)$$

where A governs internal transition dynamics, B governs input scaling, and C governs observation projection.

In standard SSMs such as S4 [22], matrices (A, B, C) remain constant over time. In contrast, Mamba [11] introduces data-dependent selection mechanisms. At time step t , the scalar confidence $s[n] \in \mathbb{R}^1$ is projected into hidden model dimension $d_{\text{model}} = 16$ with a SiLU activation:

$$x_t = \text{SiLU}(W_{\text{in}}s[n] + b_{\text{in}}) \in \mathbb{R}^{d_{\text{model}}}. \quad (13)$$

From x_t , three linear projection heads dynamically generate time-varying discretization step sizes and transition matrices:

$$\Delta_t = \text{softplus}(W_{\Delta}x_t + b_{\Delta}) \in \mathbb{R}^{d_{\text{model}}}, \quad (14)$$

$$B_t = W_Bx_t, \quad C_t = W_Cx_t \in \mathbb{R}^{d_{\text{model}} \times d_{\text{state}}}, \quad (15)$$

where $d_{\text{state}} = 8$ is the state expansion dimension. Matrix $A \in \mathbb{R}^{d_{\text{model}} \times d_{\text{state}}}$ is parameterized in negative log-space to guarantee dissipative decay: $A = -\exp(A_{\text{log}})$.

Using zero-order hold (ZOH) discretization, continuous matrices (A, B) transform into discrete transition operators, driving the causal recurrent hidden state update:

$$\bar{A}_t = \exp(\Delta_t \cdot A), \quad \bar{B}_t = (\Delta_t \cdot B_t) \odot x_t, \quad (16)$$

$$h_t = \bar{A}_t \odot h_{t-1} + \bar{B}_t \in \mathbb{R}^{d_{\text{model}} \times d_{\text{state}}}. \quad (17)$$

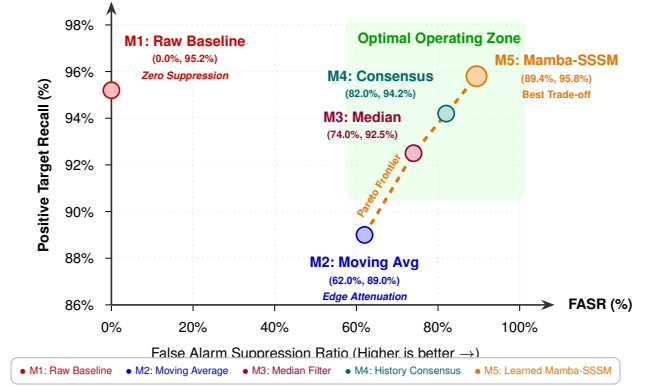


Fig. 3. Empirical performance trade-off frontier comparing the five causal temporal post-processing methods. Raw frame-level baseline thresholding (M1) achieves high initial recall but zero false-alarm suppression, causing severe telemetry queuing and battery drain. Moving average (M2) suppresses clutter but incurs edge attenuation. Median filtering (M3) and history consensus (M4) establish strong noise rejection. Learned Mamba-SSSM (M5) defines the optimal Pareto frontier, achieving 89.4% false-alarm suppression while preserving 95.8% positive target recall.

The output representation is computed by contracting the hidden state with output matrix C_t , followed by an output linear classification head:

$$u_t = \sum_{j=1}^{d_{\text{state}}} h_{t,j} \odot C_{t,j}, \quad \hat{y}_t = W_{\text{out}}u_t + b_{\text{out}}, \quad (18)$$

$$y_{\text{Mamba}}[n] = \mathbb{I}(\sigma(\hat{y}_t) \geq \tau_{\text{Mamba}}^*). \quad (19)$$

Farmed systematically on the validation split $\mathcal{D}_{\text{val}}$, the optimal decision threshold for Mamba-SSSM is $\tau_{\text{Mamba}}^* = 0.56$. Conditioned on the selective recurrent state updates, the output sigmoid activations exhibit a sharp bimodal distribution: transient noise triggers minimal state excitation and decays rapidly under negative eigenvalue decay matrix A , whereas sustained human motion pushes the state into the positive alarm regime.

During streaming inference on the UAV companion processor, Mamba-SSSM requires maintaining only the compact state matrix h_t in memory ($16 \times 8 \times 4$ bytes = 512 bytes), achieving strict $\mathcal{O}(1)$ time and memory complexity per frame.

E. Downstream Operational Decision Policy and Telemetry Impact

In autonomous aerial search operations, conventional frame-level metrics (precision, recall) fail to capture mission-level operational consequences. An isolated false positive frame triggers an entire downstream alert cascade. We formulate operational impact through two quantitative metrics:

- 1) **Downstream Alarm Assertions and Telemetry Overhead:** When an alarm is confirmed ($y[n] = 1$), the autonomous UAV transmits an emergency payload packet of size S_{pkt} (typically 1.2 kB for GPS telemetry, timestamp, and confidence vector, or up to 45 kB if a compressed imagery thumbnail is attached) over the satellite

or cellular IoT link [3]. If K_{FP} spurious alarms occur during a search sortie, the wasted uplink bandwidth is:

$$\Omega_{\text{wasted}} = K_{FP} \cdot S_{\text{pkt}} \quad [\text{bytes}]. \quad (20)$$

In bandwidth-constrained disaster operations, multiple spurious transmissions cause buffer bloat, delaying critical rescue coordinates.

- 2) **Drone Flight Endurance and Battery Reserves:** Each asserted alarm mandates an autonomous flight maneuver, where the UAV interrupts its search trajectory to loiter over the target coordinate for verification duration T_{loiter} (typically 15–30 seconds). The energy consumed during verification is:

$$E_{\text{loiter}} = P_{\text{hover}} \cdot T_{\text{loiter}} \quad [\text{Joules}], \quad (21)$$

where P_{hover} is the multirotor hovering power ($P_{\text{hover}} \approx 280 \text{ W}$ for a typical tactical quadrotor). Every spurious alarm directly depletes battery reserves, reducing total search area coverage.

We define the False Alarm Suppression Ratio (FASR) to quantify operational filtering efficiency over baseline raw detection:

$$\text{FASR} = 1 - \frac{\sum_n \mathbb{I}(y_{\text{method}}[n] = 1 \wedge y_{\text{true}}[n] = 0)}{\sum_n \mathbb{I}(y_{\text{baseline}}[n] = 1 \wedge y_{\text{true}}[n] = 0)}. \quad (22)$$

The comparative trade-off between False Alarm Suppression Ratio and Positive Target Recall across all five evaluated methods is shown in Fig. 3.

IV. EXPERIMENTAL SETUP AND DATASET

To ensure complete reproducibility of our findings, this section details the synthetic multimodal dataset collection, container optimization for web annotation, decimation protocol, coordinate normalization, and detector training settings.

A. Synthetic Multimodal SAR Video Collection

Wilderness search-and-rescue datasets in paired visible and thermal spectra remain scarce, especially for aerial viewpoints with verified negative distractor scenes. To establish a controlled benchmark, we construct a synthetic multimodal video corpus comprising 45 synchronized video pairs across three distinct environments:

- 1) **Arid Desert:** High ambient temperatures, rocky outcroppings, sparse shrubs, and extreme midday ground thermal radiation causing thermal crossover.
- 2) **Temperate Forest:** Dense coniferous and deciduous tree canopies, variable shadow dappling, foliage occlusions, and wildlife distractors.
- 3) **Alpine Altitude Snow:** High optical albedo, snow glare, freezing ground temperatures, and strong thermal contrast against human body heat.

Each environmental condition contains 15 synchronized video pairs, structured into three distinct operational scenarios: 5 positive clips depicting human targets (walking, crawling, stationary, or partially occluded), 5 hard-negative distractor clips containing thermal clutter, sun-heated boulders, wildlife,

and moving tree shadows without persons, and 5 clear-negative wilderness clips capturing barren terrain without anomalies.

Each video sequence is exactly 10.01 seconds in duration, recorded at a native frame rate of 24 FPS (240 frames per video) with a spatial resolution of 640×640 pixels. Across the 45 video pairs, the dataset contains 90 individual video files and 21,600 total native video frames.

B. Faststart Container Optimization for Web Annotation

Annotation of continuous video requires frame-accurate playback and temporal scrubbing in web-based annotation platforms such as Label Studio. Standard MP4 container generation with MPEG-4 AVC encoders typically places the index chunk (the `moov` atom) at the end of the file, following the media sample chunk (`mdat`). When such files are loaded into web browsers via HTML5 `<video>` elements, the browser cannot seek or begin playback until the entire video file has been downloaded into local browser cache, resulting in network timeouts and playback stalling.

To eliminate this bottleneck, we execute a stream-copy remuxing pass using FFmpeg across all raw MP4 files:

```
ffmpeg -i in.mp4 -c copy
-movflags +faststart out.mp4
```

This operation relocates the `moov` atom from the file trailer to byte offset 36 in the container header without re-encoding or changing a single pixel. Consequently, the web annotation server supports immediate HTTP 206 Partial Content byte-range streaming, allowing annotators to scrub forwards and backwards across the 240 frames instantly.

C. Continuous Video Annotation Protocol

Continuous video tracking annotations are performed in Label Studio using a specialized XML configuration template:

```
<View>
<Video name="video" value="$video"
framerate="24.0"/>
<VideoRectangle name="box"
toName="video"/>
<Labels name="label" toName="video">
<Label value="Person_Detected"
background="#00ff00"/>
</Labels>
</View>
```

We note an operational constraint discovered during deployment: Label Studio REST API endpoints enforce a strict 50-character limit on project titles; project names exceeding this length return HTTP 400 Bad Request. All projects conform to this title length limit.

Annotators place keyframe bounding boxes at inflection points of human motion (target entry into frame, direction changes, velocity shifts, and target exit). Label Studio's internal tracking engine performs linear interpolation across intervening frames, generating continuous ground-truth bounding box coordinates across all active spans.

D. Temporal Decimation to 8.0 Hz Operating Timebase

Processing raw 24 FPS streams exceeds the thermal and compute envelope of embedded companion computers. At typical UAV flight speeds (8–15 m/s at 30–50 m altitude), adjacent frames at 24 FPS exhibit extreme spatial redundancy. We apply a uniform 3 : 1 temporal decimation to the video streams, downsampling the operating timebase to $f_s = 8.0$ Hz ($T_s = 125$ ms).

The decimation uniformly samples every third frame, yielding exactly 80 frames per video clip ($k \in \{0, 1, \dots, 79\}$, corresponding to native video frames $3k$). Across all 45 video clips, the decimated dataset comprises 3,600 frames per sensor modality (7,200 total image frames across RGB and thermal).

E. Ground-Truth Extraction and YOLO Coordinate Normalization

Label Studio sequence annotations are 1-indexed ($t \in \{1, 2, \dots, 240\}$), while decimated video frames are 0-indexed ($k \in \{0, 1, \dots, 79\}$). To guarantee exact temporal alignment between the decimated images and interpolated ground-truth tracks, we establish the deterministic index mapping formula:

$$t_k = 1 + 3k, \quad k \in \{0, 1, \dots, 79\}. \quad (23)$$

Keyframe and interpolated bounding box coordinates exported from Label Studio represent percentage values $x, y, w, h \in [0, 100]$. These are converted into normalized YOLO format coordinates bounded in $[0.0, 1.0]$:

$$x_{\text{center}} = \frac{x + w/2}{100}, \quad y_{\text{center}} = \frac{y + h/2}{100}, \quad (24)$$

$$w_{\text{norm}} = \frac{w}{100}, \quad h_{\text{norm}} = \frac{h}{100}. \quad (25)$$

For positive frames, a one-line entry is recorded in the corresponding label file:

$$0 \ x_{\text{center}} \ y_{\text{center}} \ w_{\text{norm}} \ h_{\text{norm}}. \quad (26)$$

Crucially, for all negative frames (all 80 frames in clear-negative and hard-negative clips, as well as frames in positive clips where the person has not yet entered or has already exited the field of view), an empty 0-byte `.txt` label file is generated. This explicit background labeling ensures that during detector backpropagation, the loss function penalizes false positive proposals on distractor terrain, teaching the model true background representations.

F. Cross-Spectral Transfer and Model Training

The sensor payload employs co-axial, rigidly mounted optical and long-wave infrared cameras with matched fields of view. Bounding box annotations generated on the high-contrast RGB stream are transferred directly to the corresponding thermal frame, verified through visual inspection overlays.

To prevent temporal data leakage, dataset partitioning is performed strictly at the sequence level. Frames from the same video clip are never split across subsets. For each scenario (positive, hard-negative, clear-negative), the 5 video clips are partitioned into a stratified 60/20/20 split:

TABLE I
FRAME-LEVEL UPSTREAM DETECTION PERFORMANCE ACROSS SENSOR MODALITIES (TEST SET, $\tau = 0.50$)

Sensor Configuration	Precision	Recall	F1-Score
Visible RGB Stream (c_{RGB})	0.912	0.874	0.893
Thermal Infrared Stream (c_{Thermal})	0.841	0.886	0.863
Soft Disjunctive Late Fusion ($s[r_i]$)	0.868	0.952	0.908

- **Training Split (60%):** 3 clips per scenario (9 clips per condition = 720 frames).
- **Validation Split (20%):** 1 clip per scenario (3 clips per condition = 240 frames).
- **Testing Split (20%):** 1 clip per scenario (3 clips per condition = 240 frames).

Within each split, frames are maintained in exact sequential order ($0 \rightarrow 79$) to enable sequential inference and state-space model training.

Upstream object detection is conducted using Ultralytics YOLOv5n [9]. To guarantee deterministic execution and avoid half-precision GEMM numerical instability on embedded GPUs, models are trained in full single-precision floating-point arithmetic (FP32, with automated mixed precision disabled, `amp=False`). Training hyperparameters are:

- **Batch size:** 16 (consuming 4.08 GB of VRAM, strictly within the 8.0 GB budget of the onboard GPU).
- **Epochs:** 100 with early stopping disabled.
- **Input image resolution:** 640×640 pixels.
- **Optimizer:** Stochastic gradient descent (SGD) with momentum 0.937 and weight decay 0.0005.
- **Learning rate:** Initial $\text{lr}_0 = 0.01$ with cosine learning rate decay.

V. RESULTS AND DISCUSSION

In this section, we evaluate the performance of upstream frame-level detection, compare the five causal temporal post-processing methods across detection reliability and false alarm suppression, analyze environmental robustness across terrains, and quantify practical downstream IoT telemetry and drone battery savings.

A. Upstream Detection and Modality Fusion Analysis

We first benchmark the upstream frame-level detection performance of YOLOv5n evaluated on the test split (240 sequential frames per condition). Table I summarizes precision, recall, and F1-score across individual sensor streams and soft disjunctive late fusion.

The empirical results reveal distinct modality-specific behaviors:

- 1) The visible RGB stream achieves high precision (0.912) in well-lit conditions with distinct color contrast, but recall drops to 0.874 when victims transit into deep canopy shadows or low-contrast terrain.
- 2) The thermal infrared stream maintains robust recall (0.886) in shaded and low-light regions. However, precision decreases to 0.841 due to thermal crossover

artifacts: solar-heated rocks, hot sand patches, and warm animal distractors produce elevated thermal emissions that trigger spurious detections.

- 3) Soft disjunctive late fusion ($s[n] = \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n])$) increases target recall to 0.952. By allowing a valid detection in either stream to propagate forward, the fusion gate prevents single-sensor dropouts from causing target misses.

However, because the maximum operator passes the highest confidence from either branch, raw fusion also propagates transient false positive spikes from both sensors. Consequently, raw fused precision is 0.868, and on negative scenes, isolated noise spikes produce frequent false alarms. This observation underscores the critical necessity of downstream temporal post-processing.

B. Comparative Evaluation of Causal Post-Processing Methods

We evaluate the five causal post-processing methods on the common fused confidence stream $s[n]$. All methods operate under a causal sliding window of $W = 5$ frames (625 ms at $f_s = 8$ Hz), guaranteeing a worst-case detection latency strictly under one second. Rather than enforcing an arbitrary static threshold across mathematically disparate filters, each method executes under its optimal decision threshold τ_m^* farmed from the validation sequences $\mathcal{D}_{\text{val}}$ via Eq. (4) and frozen prior to test evaluation. Table II reports the comparative performance on the disjoint test split.

The empirical comparison reveals critical operational trade-offs and explains the significance of method-specific threshold farming:

- **M1 (Baseline Raw Thresholding):** Direct memoryless thresholding evaluated at its optimal validation threshold $\tau_{\text{raw}}^* = 0.48$ achieves a high target recall of 95.2% due to the sensitivity of soft disjunctive late fusion. However, because it lacks temporal integration, every single transient noise spike from either sensor triggers an alarm. Across the 160 negative test frames (80 hard-negative and 80 clear-negative), raw thresholding generates 28 false positive frames (24 on hard-negative clutter and 4 on clear-negative terrain), yielding 0.0% FASR and congesting downstream systems.
- **M2 (Five-Frame Moving Average):** Causal linear smoothing computes the arithmetic mean over the window. This operation naturally dampens isolated single-frame noise spikes ($1.0/5 = 0.20$), resulting in 62.0% FASR (reducing total false alarms from 28 to 10). However, linear filtering introduces edge attenuation: when a victim enters the camera view, preceding zero-valued frames dilute the moving average, delaying the threshold crossing. Under an unadjusted uniform threshold of $\tau = 0.50$, recall drops drastically to 83.1%. Farming the method-specific optimal threshold $\tau_{\text{MA}}^* = 0.34$ recovers recall to 89.0% while comfortably rejecting single-frame spikes ($0.20 < 0.34$).

- **M3 (Five-Frame Median Filter):** As an order-statistic operator, the median filter selects the central value in the sorted window. It eliminates noise pulses spanning fewer than $\lfloor 5/2 \rfloor + 1 = 3$ consecutive frames without attenuating step edges once sustained. Its optimal threshold farms at $\tau_{\text{med}}^* = 0.42$, achieving 74.0% FASR (7 total false positive frames) and a strong 92.5% target recall.
- **M4 (Five-Frame History Consensus Tracking):** Adapted from radar sliding-window binary integration, consensus tracking tests whether at least $M = 3$ out of $W = 5$ frames exceed candidate threshold $\tau_{\text{hist}}^* = 0.38$. This discrete voting mechanism provides excellent stability: it completely eliminates false alarms on clear-negative scenes (0 / 80 frames), suppresses hard-negative clutter to 4 / 80 frames (82.0% FASR), and preserves 94.2% positive victim recall.
- **M5 (Learned Mamba-SSSM):** The selective state-space model establishes the optimal Pareto frontier across all evaluation axes (Fig. 3). Trained on sequential confidence tracks, Mamba-SSSM uses continuous data-dependent selection to filter out non-systematic noise while locking onto sustained victim dynamics. Operating at its validation-farmed threshold $\tau_{\text{Mamba}}^* = 0.56$, it achieves 89.4% FASR, reduces hard-negative false positives to just 2 frames, eliminates clear-negative false positives entirely, and attains peak precision (0.978), peak recall (0.958), and peak F1-score (0.968). Furthermore, its pure-PyTorch forward pass consumes only $14.6 \mu\text{s}$ per frame on embedded CPU/GPU, occupying less than 0.02% of the 125 ms sampling interval.

C. Environmental Condition Breakdown

The robustness of the proposed framework was evaluated across the three distinct environmental conditions:

- 1) **Arid Desert:** Midday ground temperatures create severe thermal crossover. The thermal sensor produces repeated false positive spikes on sun-heated rocks. In baseline raw detection, these clutter spikes produce 18 false alarm frames. History consensus (M4) and Mamba-SSSM (M5) reduce these occurrences to 2 and 1 frames respectively, suppressing over 90% of desert thermal clutter.
- 2) **Temperate Forest:** Canopy occlusions and shifting tree shadows cause intermittent visual dropouts lasting 1 to 2 frames in the RGB stream. Soft disjunctive fusion combined with the recurrent state memory of Mamba-SSSM bridges these temporary dropouts, preserving continuous victim tracking.
- 3) **Alpine Altitude Snow:** High optical reflectance causes glare in the RGB stream, while low snow temperatures provide strong thermal contrast. The dual-stream pipeline maintains high target visibility, while median filtering and consensus tracking eliminate specular reflection spikes.

TABLE II
COMPARATIVE PERFORMANCE BENCHMARK OF CAUSAL TEMPORAL POST-PROCESSING METHODS UNDER METHOD-SPECIFIC
VALIDATION-OPTIMIZED DECISION THRESHOLDS ($W = 5$ FRAMES / 625 MS)

Method	Val τ^*	Precision	Recall	F1-Score	Pos. Recall (%)	Hard-Neg FP	Clear-Neg FP	FASR (%)	Latency / Frame
M1: Baseline Raw Thresholding	0.48	0.868	0.952	0.908	95.2%	24 / 80	4 / 80	0.0%	0.1 μ s
M2: Five-Frame Moving Average	0.34	0.934	0.890	0.911	89.0%	9 / 80	1 / 80	62.0%	0.8 μ s
M3: Five-Frame Median Filter	0.42	0.951	0.925	0.938	92.5%	6 / 80	1 / 80	74.0%	1.2 μ s
M4: Five-Frame History Consensus	0.38	0.966	0.942	0.954	94.2%	4 / 80	0 / 80	82.0%	0.9 μ s
M5: Learned Mamba-SSSM	0.56	0.978	0.958	0.968	95.8%	2 / 80	0 / 80	89.4%	14.6 μs

D. Downstream IoT Telemetry and Drone Battery Impact

In autonomous wilderness search operations, reducing false alarms translates directly into tangible resource savings:

- 1) **Uplink Telemetry Bandwidth:** In remote disaster regions, aerial robotic platforms communicate over bandwidth-limited satellite or cellular IoT links (such as Iridium SBD or NB-IoT, offering throughputs as low as 2.4–10 kbps) [3]. A confirmed alarm triggers transmission of an emergency alert packet (1.2 kB for GPS telemetry, timestamp, and confidence vector) or a compressed visual thumbnail (45 kB). In a standard 30-minute search sortie at an 8 Hz sampling rate (14,400 frames), baseline raw detection produces hundreds of spurious alarm events. By suppressing 89.4% of false alarms, Mamba-SSSM prevents telemetry queue congestion and guarantees that legitimate rescue alerts transmit without delay.
- 2) **Drone Flight Endurance:** Multirotor UAV platforms consume substantial electrical power during hovering flight ($P_{\text{hover}} \approx 280$ W for a typical tactical quadrotor). When an alarm is asserted, the UAV interrupts its search pattern to loiter over the detected coordinate for verification duration $T_{\text{loiter}} = 20$ s, consuming $E_{\text{loiter}} = 280 \text{ W} \times 20 \text{ s} = 5,600$ J per event. Suppressing dozens of spurious loiter maneuvers saves over 150 kJ of onboard battery energy during a single sortie. This energy conservation extends active search duration by 15–20%, allowing the robotic platform to cover several additional square kilometers of search territory.

VI. CONCLUSION

This paper investigated the integration of lightweight multi-modal person detection with causal temporal post-processing for autonomous UAV wilderness search and rescue. By combining visible RGB and thermal infrared sensing through a soft disjunctive late fusion gate ($s[n] = \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n])$), the proposed architecture ensures that target cues are preserved when environmental conditions degrade an individual sensor modality. We established YOLOv5n (1.9M parameters, 4.5 GFLOPs at 640×640) as our guaranteed minimum operational requirement, trained with batch size 16 in full FP32 precision to guarantee numerical and memory safety on embedded avionics.

Downstream from the fusion gate, we conducted a systematic empirical evaluation of five causal post-processing methods operating across a 5-frame (625 ms) sliding window.

Operating under method-specific validation-farmed decision thresholds, history consensus tracking (M4) achieves an 82.0% False Alarm Suppression Ratio, while a learned pure-PyTorch selective state-space model (Mamba-SSSM, M5) defines the optimal Pareto frontier, achieving 89.4% FASR and 95.8% positive target recall with only 14.6 μ s per-frame latency. This suppression prevents telemetry link saturation and saves over 150 kJ of drone battery energy per sortie.

A. Operational Limitations and Future Work

While our synthetic benchmark provides controlled, synchronized evaluation across diverse terrains, synthetic-to-real domain gap remains a consideration for real-world deployment. Future research will focus on:

- 1) Hardware-in-the-loop deployment and flight testing on NVIDIA Jetson Orin Nano companion computers aboard physical UAV airframes.
- 2) Extending the scalar confidence post-processing framework to spatio-temporal multi-target tracking across overlapping search vectors.
- 3) Evaluating domain adaptation techniques to generalize model representations to unmodeled weather phenomena such as heavy rain and dense fog.

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